

THE ACME ELECTRONIC SUPERMARKET

MOBILE COMPUTING

GROUP 1

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01

SYSTEM OVERVIEW

Supermarket System is composed of three main components:



Used by customers for scanning products from the Supermarket, managing accounts, and completing purchases

CLIENT APP



Used by store attendants or integrated in machines to scan the QR Code transaction generated in the Client App and handle checkouts.

TERMINAL CHECKOUT APP



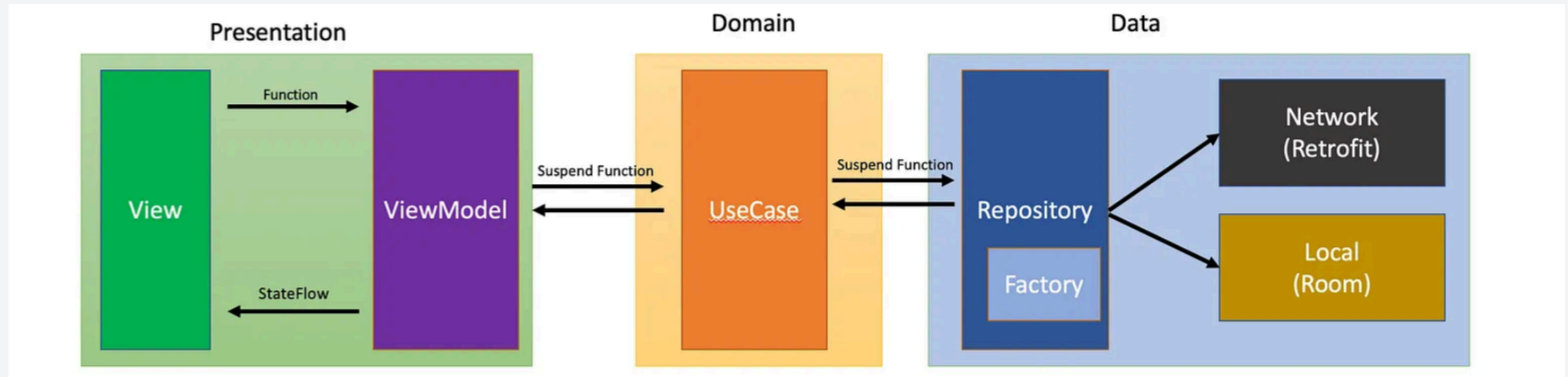
Which acts as the backend system, exposing RESTful APIs to manage users, transactions and vouchers logic.

CENTRAL SERVER

02

ARCHITECTURE

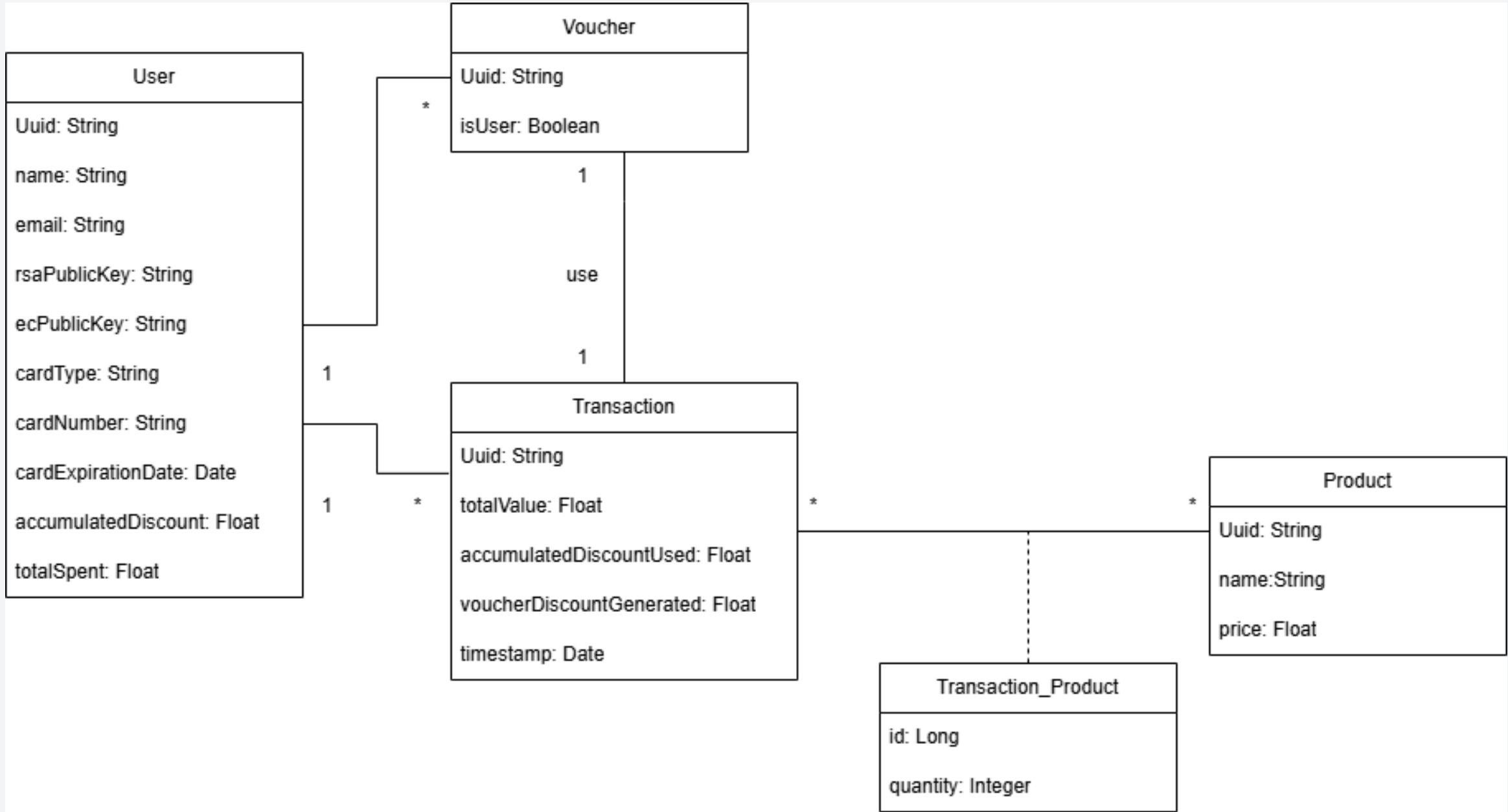
Clean Architecture and MVVM: Model-View-ViewModel



Hilt (Dependency Injection) is used to separate responsibilities (Separation of Concerns), facilitate unit testing, improve maintainability and reduce coupling between components of the architecture.

03

DATABASE SCHEMA



Room (Android)

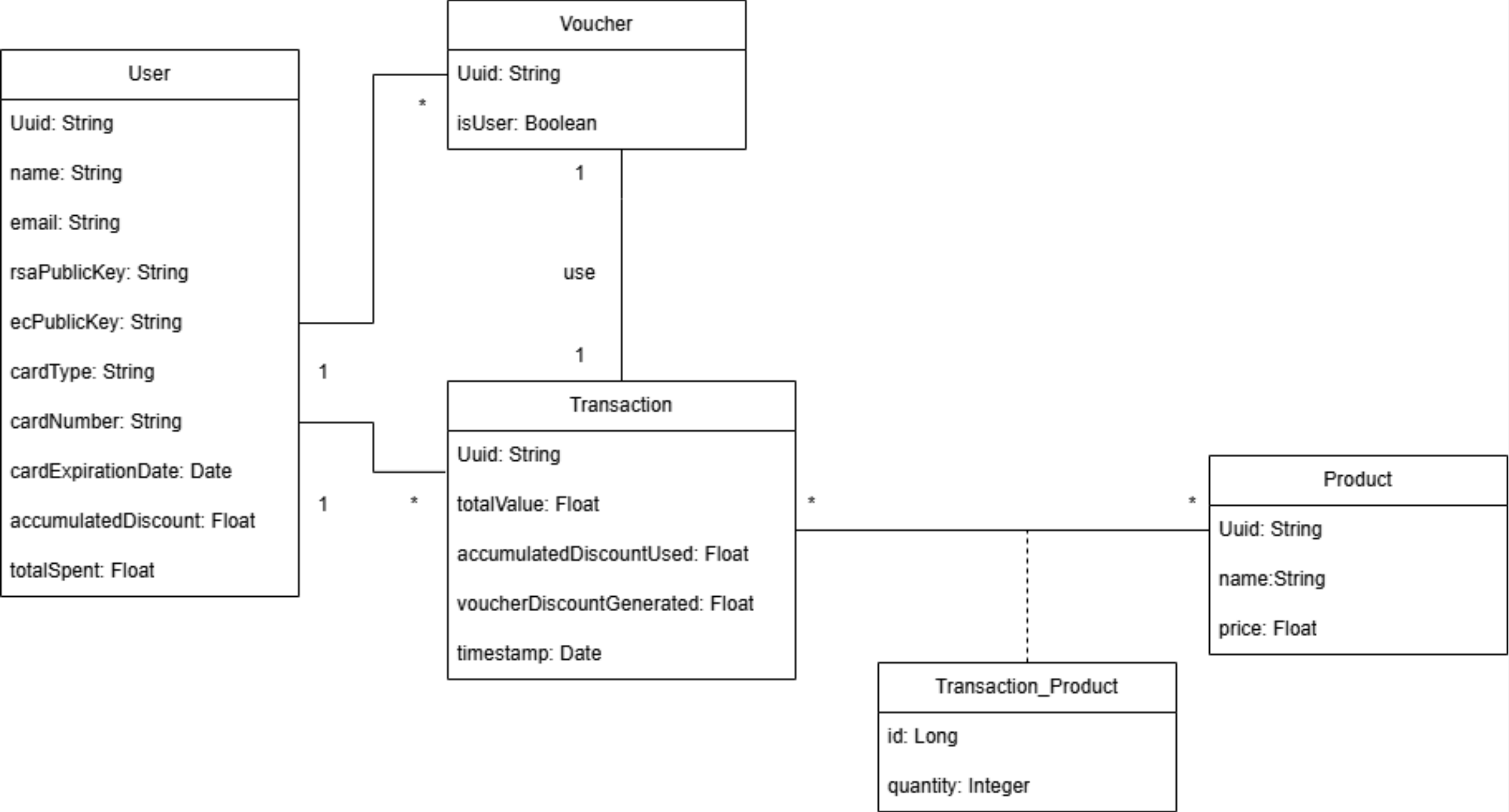
Android's local persistence library that provides a layer of abstraction on top of SQLite, allowing to store data offline efficiently.

H2 (Spring Boot)

Lightweight in-memory database, ideal for rapid backend development

03

DATABASE SCHEMA



DTOS

Defines the API contract, guaranteeing separation between layers and consistent data communication.

DAOS

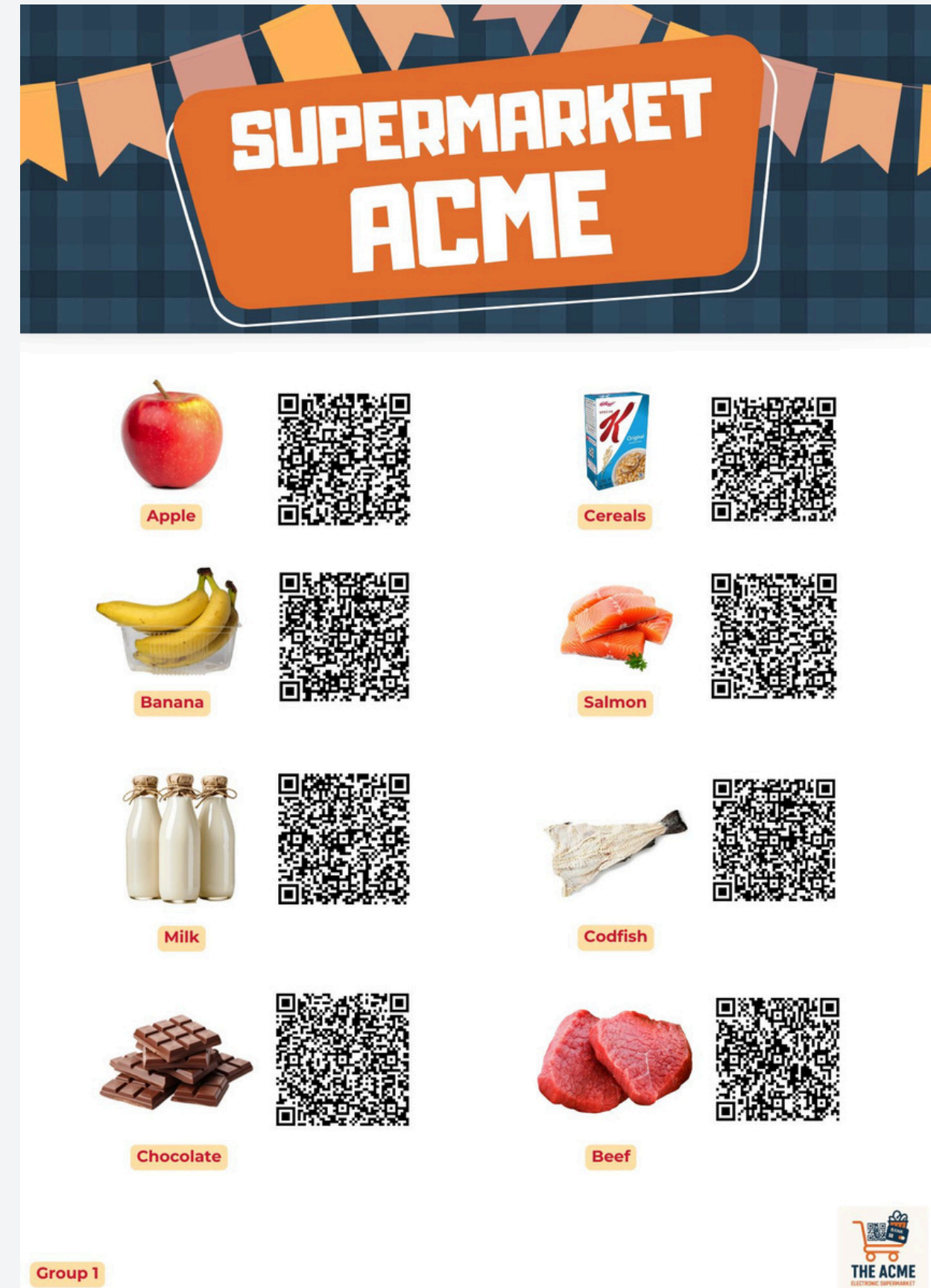
Defines how to access the local database (Room)

User Registration / Login

QR-code scanning to add items and process the checkout

QR-code generation with product list, optional voucher, and accumulated discount flag (signed with user EC private key)

Request past transactions and available vouchers (signed with user EC private key)



05

LIVE DEMO

LIVE DEMO

